Project Report

Modular Battery Management System(BMS)

1.Project title

Modular Battery Management System **(**BMS**)** using C programming**.**

2.Objective

The objective of this project is to design a Modular Battery Management System (BMS) that monitors the voltages of battery cells, identifies the strongest and weakest cells, calculates the voltage imbalance, detects whether the imbalance is increasing or decreasing over time, and applies an adaptive threshold based on the Battery State of Charge (SoC). The system is scalable and can support different numbers of battery cells.

3.Introduction

A Battery Management System (BMS) is an electronic system that monitors and manages rechargeable battery packs. It ensures safe operation, improves battery life, and prevents damage caused by voltage imbalance. This project implements a modular BMS using the C programming language.

4.Working

- Reads the voltage of all battery cells.
- Finds the strongest (highest voltage) and weakest (lowest voltage) cells.
- Calculates the voltage imbalance.
- Compares the current imbalance with the previous imbalance.
- Uses SoC to determine the adaptive threshold.
- Displays battery status as Healthy or Battery Balancing Required.

5.Scalability

The program uses:

#define NUM_CELLS 4

By changing this value to 8, 12, or 16, the same code can support more battery cells without any other modifications.

6.Memory usage

The program stores battery voltages in a float array. Memory usage increases with the number of battery cells. Therefore, the memory complexity is O(n).

7.Execution Time

The program checks each battery cell only once to find the highest and lowest voltages. Hence, the execution time is O(n).

8.Data Structures Used

- Float array (cellVoltage[])
- Float variables (highest, lowest, imbalance, soc, threshold)
- Integer variables (highestCell, lowestCell)

9.Sample Output

- Strongest Cell: Cell 3 (3.75 V)
- Weakest Cell: Cell 2 (3.68 V)
- Voltage Imbalance: 0.07 V
- Trend: Imbalance Increasing
- Adaptive Threshold: 0.03 V
- Status: WARNING! Battery Balancing Required.

10.Conclusion

The Modular Battery Management System successfully monitors battery cell voltages, identifies the strongest and weakest cells, calculates voltage imbalance, and determines battery health using an adaptive threshold. The modular design allows the program to scale easily from 4 to 16 cells, making it efficient and suitable for battery monitoring applications.